UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

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This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. 3D animation does not use modelling.
2. A bitmap image is made up of pixels.
3. A graphics tablet is used to draw directly into a computer.
4. Interlacing displays every line of a video frame at the same time.
5. Copyright protects original multimedia works.
6. Multimedia combines text, graphics, audio, video and animation.
7. RAM is a type of permanent storage.
8. Hertz measures sound frequency.
9. Streaming allows media to begin playing before the whole file is downloaded.
10. CMYK is used for printed output.
11. Tweening creates intermediate frames between keyframes.
12. A scanner is an input device.
13. Keyframes define important positions in an animation.
14. A higher sampling rate generally produces a more accurate recording.
15. Hypermedia combines multimedia with hyperlinks.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A

A. Frame rate	B. Vector	C. Bitmap	D. Linear multimedia	E. Storyboard
F. Multimedia	G. Streaming	H. Virtual reality	I. Chroma key	J. Browser
K. RGB	L. Keyframe	M. Serif	N. Prototype	O. Card-based tool

1. visual sequence plan
2. software that displays web pages
3. frames displayed per second
4. replacing a colour background
5. simulated environment
6. playing media while it downloads
7. font with small strokes at character ends
8. integration of text, graphics, audio, video and animation
9. important frame in an animation
10. image made of pixels
11. presentation without user control of sequence
12. authoring tool organised like pages or cards
13. image made of mathematical objects
14. early model of a product
15. additive colour model used by screens

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Target audience refers to:
A. Intended users
B. File format
C. Software licence
D. Colour model

2. Which is a 3D animation task?
A. Sampling
B. Kerning
C. Bookmarking
D. Modelling and rendering

3. An authoring tool is used to:
A. Only browse the web
B. Only record audio
C. Only print text
D. Assemble multimedia elements

4. Hypertext contains:
A. No text
B. Links to other content
C. Only images
D. Only audio

5. A browser is used to:
A. Scan images
B. Print documents only
C. Display web pages
D. Edit audio

6. Amplitude of a sound wave relates to:
A. Pitch
B. Loudness
C. Frame rate
D. Colour

7. Which is an input device?
A. Monitor
B. Plotter
C. Scanner
D. Printer

8. Interactivity in multimedia means:
A. The user can control content and flow
B. Only text is shown
C. There is no navigation
D. The system plays automatically

9. RAM is:
A. A web browser
B. Temporary working memory
C. An input device
D. Permanent storage

10. Which is an audio file format?
A. JPEG
B. MPEG
C. HTML
D. WAV

11. A storyboard is:
A. A visual sequence plan
B. A sound file
C. A video codec
D. A colour model

12. Which colour model is subtractive?
A. CMYK
B. RGB
C. HSL
D. HSV

13. Kerning adjusts:
A. Space between individual character pairs
B. Page size
C. Font colour
D. Line length

14. A bitmap image is made of:
A. Pixels
B. Text characters
C. Sound samples
D. Mathematical objects

15. Which best defines multimedia?
A. Text only
B. A single image
C. Text, graphics, audio, video and animation
D. A printed book

16. Morphing changes:
A. One shape into another
B. File format
C. Audio pitch
D. Font size

17. Cel animation is created by:
A. Drawing each frame by hand
B. Scanning text
C. Recording sound
D. Printing images

18. Which is an authoring tool category?
A. RGB
B. CMYK
C. Card-based
D. JPEG

19. Hertz measures:
A. File size
B. Frequency
C. Loudness
D. Resolution

20. Which is an output device?
A. Scanner
B. Mouse
C. Keyboard
D. Plotter

21. Which is NOT a multimedia element?
A. Animation
B. Text
C. Compiler
D. Audio

22. Which is a benefit of multimedia in education?
A. It combines visuals and sound
B. It replaces all teachers
C. It removes all interaction
D. It uses only text

23. Convergence in multimedia refers to:
A. Merging of media and computing
B. Deleting media files
C. Printing images
D. Recording audio only

24. CMYK is used mainly for:
A. Printed output
B. Screen display
C. Sound mixing
D. Animation

25. MPEG is:
A. A video compression family
B. A storage device
C. A font
D. An image editor

26. A prototype is:
A. An early model of a product
B. A final release
C. A codec
D. A font

27. Hypermedia is best described as:
A. Text-only media
B. Multimedia with hyperlinks
C. A single video file
D. A printed manual

28. Streaming allows media to:
A. Play while downloading
B. Require full download first
C. Only play offline
D. Remove compression

29. MIDI stores:
A. Video frames
B. Musical notes and instructions
C. Digitized audio samples
D. Images

30. A hyperlink links to:
A. Only audio
B. Only text
C. Only images
D. Another resource

PART D (40 MARKS)

QUESTION 1

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and

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HTML context.

(4 marks)

QUESTION 2

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(5 marks)

QUESTION 3

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 4

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

END OF QUESTION PAPER